Robust & Stochastic Control

MEAM 5170

2026 September 29

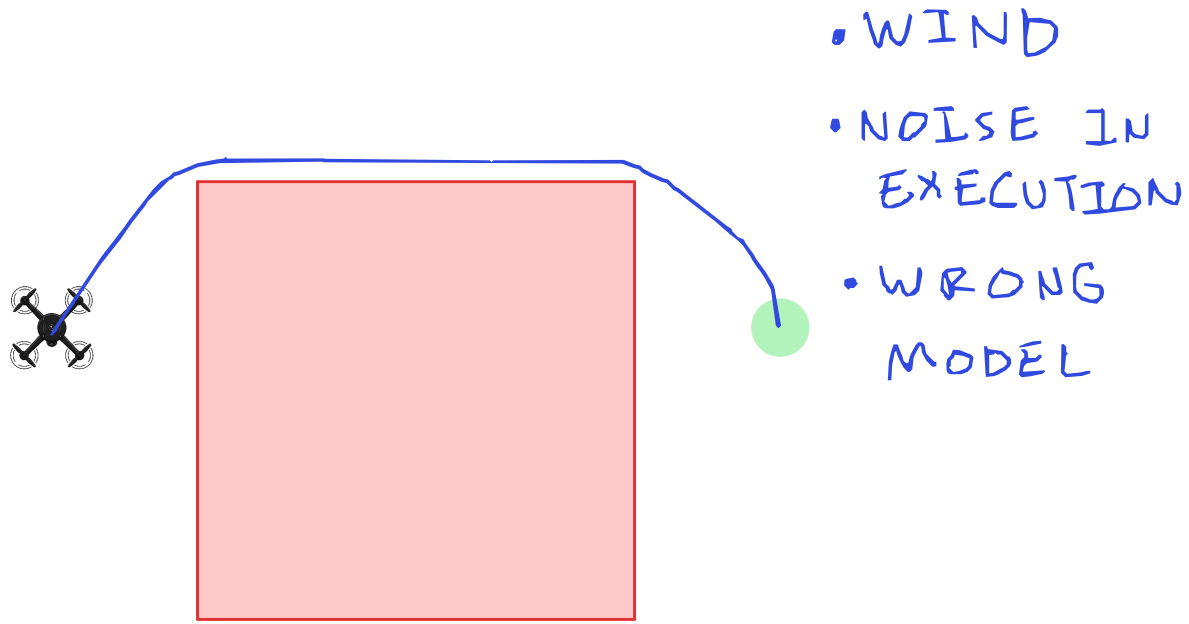
Last Time: Trajectory Optimization

Here is a general form of discrete-time trajectory optimization:

$$\begin{aligned} \min_{x_{0:T}, u_{0:T-1}} \quad & c_T(x_T) + \sum_{t=0}^{T-1} c(x_t, u_t) \\ \text{s.t. } & x_{t+1} = f(x_t, u_t), \quad x_0 = x_{\text{init}}, \\ & x_t \in \mathcal{X} \\ & u_t \in \mathcal{U} \end{aligned}$$

A Quadrotor Example

Say we want to find a minimum length trajectory for a quadrotor to get from point A to point B:



Question: *what could go wrong?*

Defining a Stochastic Control System

Previously:

$$x_{t+1} = f(x_t, u_t), \qquad x_0 = x_{\text{init}}$$

Now:

$$x_{t+1} = f(x_t, u_t, w_t), \qquad x_0 \sim p(x_0),$$

where w_t is a random process.

NOTE x_t ARE RANDOM VARIABLES

- DISTURBANCES
- MODEL UNCERTAINTIES

Continuous vs Discrete

We can talk about stochastic systems in either continuous or discrete time. For simplicity, this lecture mostly focuses on discrete time.

In continuous time, you deal with **stochastic differential equations**[1], and have to worry about Itô's lemma, which describes how the chain rule works for SDE's.

[1] Oksendal, Bernt. Stochastic differential equations: an introduction with applications. Springer Science & Business Media, 2013.

Example: Stochastic LTI

Consider:

$$x_{t+1} = Ax_t + Bu_t + w_t, \qquad w_t \sim \mathcal{N}(0, \Sigma)$$

For some noise covariance $\Sigma = \Sigma^\top \succ 0$

Note: *if x_0 is a Gaussian and u_t is an affine function of x_t, then all x_t are Gaussian.*

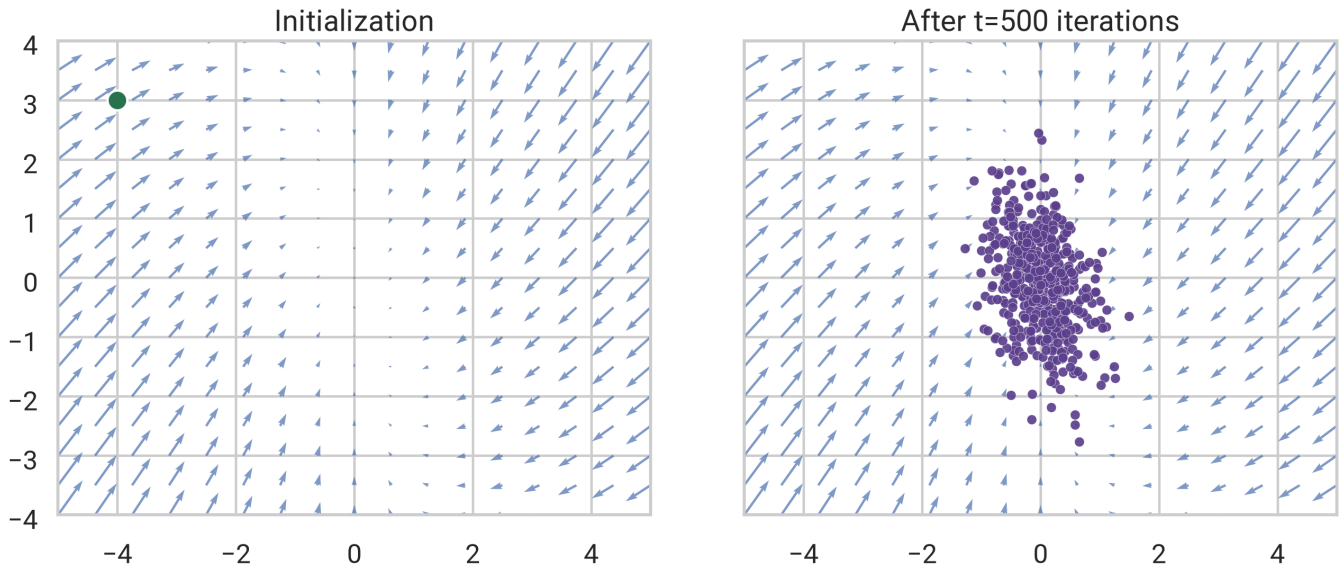
Let $u_t = Kx_t$ such that $\rho(A + BK) < 1$.

Question: Does the system have a fixed point at $x = 0$?

NO

Stationary Distributions

For stochastic systems, it often makes more sense to talk about a *stationary distribution* as opposed to a stationary point.



Question: *For the SLTI system, if $x_0 = 0$, what happens to x_t as $t \to \infty$?*

$$x_{t+1} = (A + BK)x_t + w_t, \qquad w_t \sim \mathcal{N}(0, \Sigma) \text{ I.I.D.}$$

Example: Polytope Uncertainty and LTI

Consider:

$$x_{t+1} = (A + BK)x_t + w_t, \qquad w_t \in \mathcal{W},$$

where $\mathcal{W} \subset \mathbb{R}^n$ is a convex polytope and $x_0 = 0$.

The *Minkowski sum* of two sets is

$$\mathcal{X} \oplus \mathcal{W} = \{x + w : x \in \mathcal{X}\ w \in \mathcal{W}\}$$

Also define $A\mathcal{X} = \{Ax : x \in \mathcal{X}\}$ for matrix A and set $\mathcal{X}$.

Then our set-valued dynamics are:

$$\mathcal{X}_{t+1} = (A + BK)\mathcal{X}_t \oplus \mathcal{W}$$

We can track the convex polytope $\mathcal{X}_t$ by its vertices $\{v_x^{(1)}, \ldots, v_x^{(m)}\}$

Zonotopes

Things are easier if our sets are *Zonotopes*[2], defined:

$$\mathcal{Z}(c, G) = \left\{ c + \sum_{i=1}^{m} z_i g_i : z_i \in [-1, 1] \right\}$$

Then, the Minkowski sum becomes:

$$\mathcal{Z}(c, G) \oplus \mathcal{Z}(c', G') = \mathcal{Z}(c + c', [G,\ G'])$$

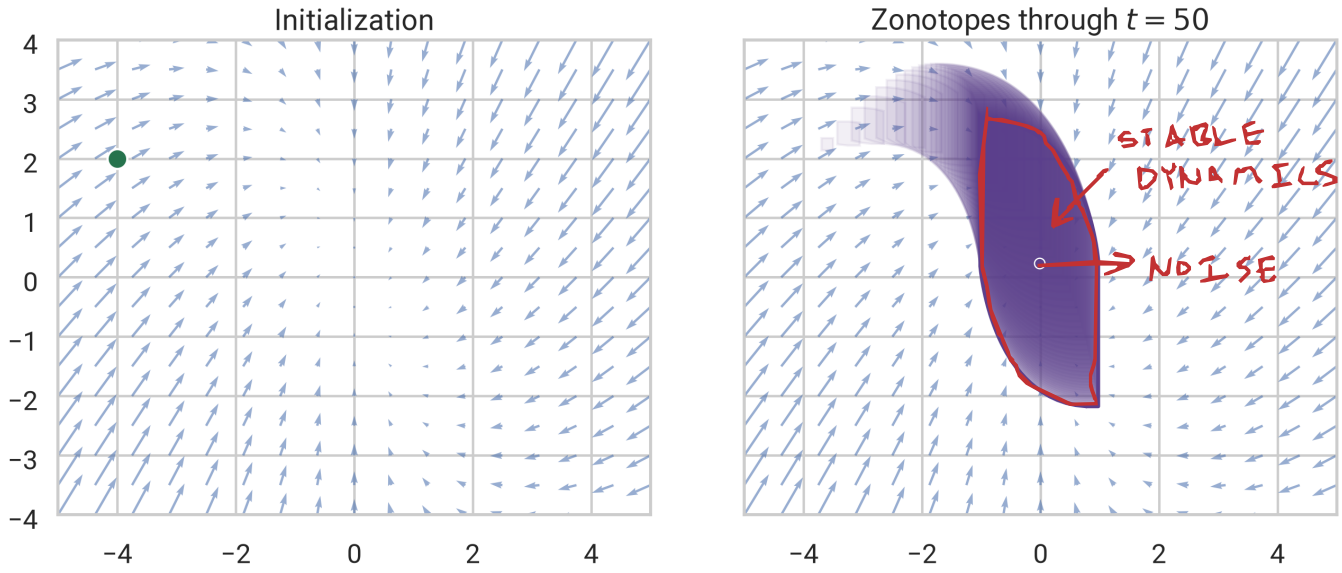
So, if $\mathcal{X}_t = \mathcal{Z}(c_x, G_x)$ and $\mathcal{W} = \mathcal{Z}(c_w, G_w)$, we get:

$$\mathcal{X}_{t+1} = \mathcal{Z}\Big((A + BK)c_x + c_w, \big[(A + BK)G_x,\ G_w\big] \Big)$$

$O(TM)$

[2] Girard, Antoine. "Reachability of uncertain linear systems using zonotopes." HSCC. Vol. 3414. 2005.

Example: Zonotope evolution



Takeaway: *We can think about propagating sets through dynamics to model uncertainty*

Example: Manipulator Equations with Uncertainty

Recall:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = Bu$$

We might have uncertainty about physical parameters (e.g. mass, friction). We can represent those by ξ:

$$M(q, \xi)\ddot{q} + C(q, \dot{q}, \xi)\dot{q} + g(q, \xi) = Bu$$

Then we would get the system:

$$\dot{x} = \begin{bmatrix} \dot{q} \\ \ddot{q} \end{bmatrix} = \begin{bmatrix} \dot{q} \\ M(q, \xi)^{-1} \left(Bu - C(q, \dot{q}, \xi)\dot{q} - g(q, \xi) \right) \end{bmatrix}$$

where our random process, $w_t = \xi$, is a random constant (over time).

Disturbances vs Uncertainty about Parameters

Stochastic system:

$$x_{t+1} = f(x_t, u_t, w_t), \qquad x_0 \sim p(x_0),$$

We might want to separate random parameters ξ that remain constant over time:

$$x_{t+1} = f(x_t, u_t, w_t, \xi), \qquad x_0 \sim p(x_0)$$

where ξ is a random variable

Note: *Random parameters can be wrapped into the state by creating an augmented state (but then you might lose full observability)*

$$\begin{bmatrix} x_{t+1} \\ \xi_{t+1} \end{bmatrix} = \begin{bmatrix} f(x_t, u_t, \xi_t) \\ \xi_t \end{bmatrix} \qquad \begin{bmatrix} x_0 \\ \xi_0 \end{bmatrix} \sim \begin{bmatrix} p(x_0) \\ p(\xi) \end{bmatrix}$$

Stochastic and Robust Optimal Control

$$x_{t+1} = f(x_t, u_t, w_t), \qquad x_0 \sim p(x_0),$$

Let $\pi \in \Pi$ be a (possibly time-varying) policy (for trajectory optimization, this is parameterized by the inputs $u_0, u_1, \ldots, u_{T-1}$).

Let $J(\pi; w_{0:T-1})$ be a cost function over policies based on $w_{0:T-1}$.

Question: *What objective should we optimize?*

Average case?

$$\min_{\pi \in \Pi} \mathbb{E}_w J(\pi; w_{0:T-1})$$

Worst case?

$$\min_{\pi \in \Pi} \max_{w_{0:T-1} \in \mathcal{W}} J(\pi; w_{0:T-1})$$

Ex: Stoch. Trajectory Optimization via Sampling

Consider a stochastic trajectory optimization problem:

$$\min_{X_{0:T}, u_{0:T-1}, K} \mathbb{E}\left[c_T(X_T) + \sum_{t=0}^{T-1} c(X_t, u_t)\right]$$
$$\text{s.t. } X_{t+1} = f(X_t, u_t, W_t), \quad X_0 = x_{\text{init}},$$
$$u_t \in \mathcal{U} \qquad u_t = K x_t$$

An easy way to approximate this is to sample $\left\{w_t^{(i)}\right\}$ and solve:

$$\min_{x_{0:T}^{(i)}, u_{0:T-1}, K} \frac{1}{n} \sum_{i=1}^{n} \left[c_T(x_T^{(i)}) + \sum_{t=0}^{T-1} c(x_t^{(i)}, u_t)\right]$$
$$\text{s.t. } x_{t+1}^{(i)} = f(x_t^{(i)}, u_t, w_t^{(i)}), \quad x_0^{(i)} = x_{\text{init}}$$
$$u_t \in \mathcal{U} \qquad u_t = K x_t$$

Question: *why might sharing the u_t between samples be conservative?*

Example: Robust TO with Finite Uncertainty Set

If our uncertainty is drawn from a finite set, $\{\xi^{(1)}, \ldots, \xi^{(n)}\}$, then we can do a similar thing:

$$\min_{x_{0:T}^{(i)}, u_{0:T-1}} \max_{i \in 1,\ldots,n} \left\{ c_T(x_T^{(i)}) + \sum_{t=0}^{T-1} c(x_t^{(i)}, u_t) \right\}$$
$$\text{s.t. } x_{t+1}^{(i)} = f(x_t^{(i)}, u_t, \xi^{(i)}), \quad x_0^{(i)} = x_{\text{init}}$$

Question: *why doesn't this work for sampling from an uncertainty set?*

WE W

SAMPLE FROM UNIF(W)

Example: Tube Trajectory Optimization

Consider a bounded uncertainty set $\mathcal{W}$ and linearized dynamics around a nominal trajectory with error feedback.

We want to optimize a *tube* around it so that we are guaranteed to remain in the tube[3]. We pick an template set $\mathcal{E}$ and can formulate:

$$\min_{\bar{x}_{0:T}, \bar{u}_{0:T-1}, \alpha_{0:T}} \; c_T(\bar{x}_T) + \sum_{t=0}^{T-1} c(\bar{x}_t, \bar{u}_t) + \sum_{t=1}^{T} \gamma \alpha_t$$

$$\text{s.t. } \bar{x}_{t+1} = A_t \bar{x}_t + B_t \bar{u}_t + d_t \qquad \bar{x}_0 = x_{\text{init}}$$

$$\alpha_{t+1} \mathcal{E} \supseteq (A_t + B_t K_t)(\alpha_t \mathcal{E}) \oplus \mathcal{W} \quad \alpha_t \geq 0$$

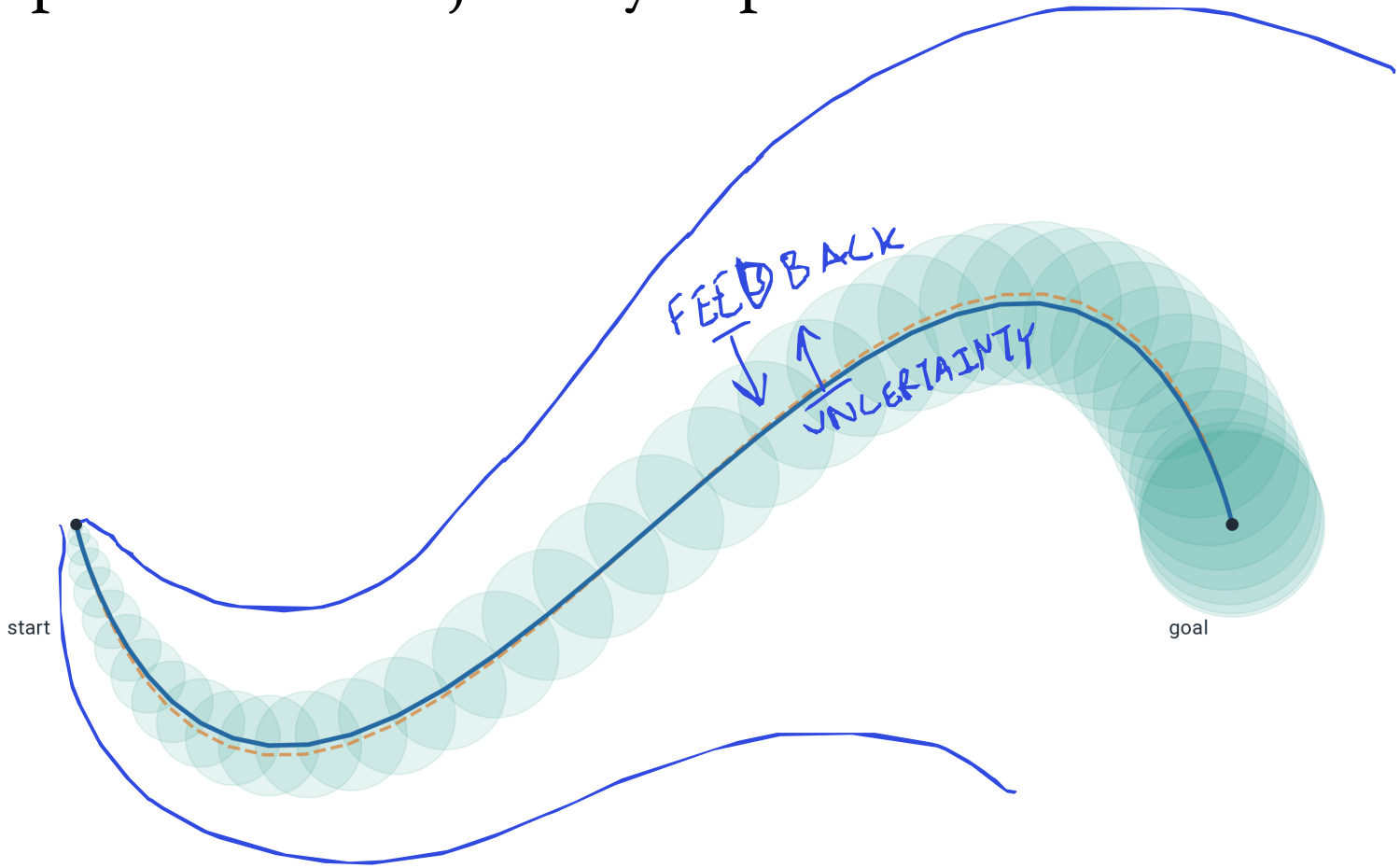
$$\bar{x}_t + \alpha_t \mathcal{E} \subseteq \mathcal{X}, \quad \bar{u}_t + K_t(\alpha_t \mathcal{E}) \subseteq \mathcal{U}$$

If $\mathcal{E} = \{x \in \mathbb{R}^d : \|x\| \leq 1\}$ and $\mathcal{W}$ has radius r_w, the tube dynamics constraint becomes:

$$\alpha_{t+1} \geq \|A_t + B_t K_t\|_2 \alpha_t + r_w$$

[3] Raković, Saša V., et al. "Homothetic tube model predictive control." Automatica 48.8 (2012): 1631-1638.

Example: Tube Trajectory Optimization



Note: *Under deterministic dynamics and initialization, feedback during trajectory optimization has no effect, but under uncertainty, feedback matters.*

Chance Constraints

We don't always just care about the objective; we may also want *constraints* to be satisfied (think about the quadrotor).

However, a hard constraint like:

$$g(x_t, u_t) \leq 0$$

might not be feasible in stochastic systems (e.g. Gaussian noise)

Instead, we can express a *chance constraint*[4] as:

$$P\left[g(x_t, u_t, w_t) \leq 0\right] \geq 1 - \alpha$$

[4] Charnes, Abraham, and William W. Cooper. "Chance-constrained programming." Management science 6.1 (1959): 73-79.

How to Actually Solve Chance Constraints

Consider TO with a linear chance constraint for $a^\top x + b^\top u \leq c$:

$$\min_{X_{0:T}, u_{0:T-1}} \quad \mathbb{E}\left[c_T(X_T) + \sum c(X_t, u_t)\right]$$

$$\text{s.t. } X_{t+1} = f(X_t, u_t, W_t), \quad X_0 = x_{\text{init}}$$

$$P\left[a^\top X_t + b^\top u_t \leq c\right] \geq 1 - \alpha$$

For linear dynamics and Gaussian distributions, the constraint is:

$$a^\top \mu_{X_t} + \underbrace{\Phi^{-1}(1-\alpha)}_{\text{INVERSE CDF}} \sqrt{a^\top \Sigma_{X_t} a} + b^\top u_t \leq c$$

We can sample to solve certain chance constraints for small α:

$$g(x_t^{(i)}, u_t, w_t^{(i)}) \leq 0, \quad \forall i, t$$

Need M^* samples, for $1 - \delta$ prob of constraint satisfaction.[5]

[5] Calafiore, Giuseppe Carlo, and Marco C. Campi. "The scenario approach to robust control design." IEEE Transactions on automatic control 51.5 (2006): 742-753.

Stochastic Value Function

Let $w_0, w_1, \ldots$ be IID from $p(w)$.

Consider cost $c(x, u, w)$.

The cost-to-go or value function is:[6]

$$
\begin{aligned}
V(x) &= \min_{\pi} \mathbb{E}_{w_{0:\infty}} \left[\sum_{t=0}^{\infty} \gamma^t c(x_t, u_t, w_t) \right] \\
&= \min_{u} \mathbb{E}_{w} \left[c(x, u, w) + \gamma \min_{\pi} \mathbb{E}_{w_{1:\infty}} \left[\sum_{t=1}^{\infty} \gamma^{t-1} c(x_t, u_t, w_t) \right] \right] \\
&= \min_{u} \mathbb{E}_{w} \Big[c(x, u, w) + \gamma V(f(x, u, w)) \Big]
\end{aligned}
$$

Question: *why did we assume IID disturbance?*

[6] Bertsekas, Dimitri P., et al. Dynamic programming and optimal control. Vol. 1. No. 2. Belmont, MA: Athena scientific, 1995.

Robust Value Function

Let $(w_0, w_1, ...) \in \mathcal{W} \times \mathcal{W} \times ...$; this is a *rectangular* uncertainty set.

With a cost $c(x, u, w)$, we can similarly derive a Bellman equation:

$$V(x) = \min_u \max_w \left\{ c(x, u, w) + \gamma V(f(x, u, w)) \right\}$$

Example: Stochastic LQR [7]

$$x_{t+1} = Ax_t + Bu_t + w_t, \qquad w_t \sim \mathcal{N}(0, \Sigma)$$

Consider the cost:

$$c(x, u) = x^\top Q x + u^\top R u,$$

for $Q = Q^\top \succeq 0$ and $R = R^\top \succ 0$.

Then, our value function is:

$$V(x) = \min_u \mathbb{E}_w \left[x^\top Q x + u^\top R u + \gamma V(\underbrace{Ax + Bu + w}_{\text{DYNAMICS}}) \right]$$

[7] Anderson, Brian DO, and John B. Moore. Optimal control: linear quadratic methods. Courier Corporation, 2007.

Example: Stochastic LQR

Let's assume our optimal policy is $u_t = Kx_t$

and our value function is $V(x) = x^\top Sx + c$ for some $S = S^\top \succeq 0$ and c.

Then, we get:

$$S = Q + \gamma A^\top SA - \gamma^2 A^\top SB(R + \gamma B^\top SB)^{-1} B^\top SA$$

$$c = \frac{\gamma}{1-\gamma} \mathbb{E}_w[w^\top Sw] = \frac{\gamma}{1-\gamma} \text{tr}(S\Sigma)$$

Question: *what happens if* $\gamma = 1$*?*

$$\lim_{T \to \infty} \frac{1}{T} \mathbb{E}_w \left[\sum_{t=0}^{T} C(x, u, w) \right]$$

Observability

Our current formulation,

$$x_{t+1} = f(x_t, u_t, w_t)$$

doesn't give us uncertainty about state from partial-observability.
This requires introducing observations:

$$y_t = h(x_t, u_t, v_t)$$

where v_t is a stochastic process handling measurement noise.

Example: LQG

$$x_{t+1} = Ax_t + Bu_t + w_t, \qquad w_t \sim \mathcal{N}(0, \Sigma)$$

We can extend our stoch. LTI system to have uncertain observations:

$$y_t = Cx_t + v_t, \qquad v_t \sim \mathcal{N}(0, \Omega)$$

Then, we arrive at a *Kalman filter*.

Adding a Kalman filter in LQR gives *LQG*, where we track *belief* $(\hat{x}, P)$:

$$\begin{aligned}
\hat{x}_{t|t-1} &= A\hat{x}_{t-1|t-1} + Bu_{t-1}, \\
P_{t|t-1} &= AP_{t-1|t-1}A^\top + \Sigma, \\
L_t &= P_{t|t-1}C^\top \left(CP_{t|t-1}C^\top + \Omega\right)^{-1}, \\
\hat{x}_{t|t} &= \hat{x}_{t|t-1} + L_t \left(y_t - C\hat{x}_{t|t-1}\right), \\
P_{t|t} &= (I - L_t C)P_{t|t-1}, \\
u_t &= K^* \hat{x}_{t|t}.
\end{aligned}$$

The Margins of LQG[8]

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Guaranteed Margins for LQG Regulators

JOHN C. DOYLE

***Abstract*—There are none.**

Introduction

Considerable attention has been given lately to the issue of robustness of linear–quadratic (LQ) regulators. The recent work by Safonov and Athans [1] has extended to the multivariable case the now well-known guarantee of 60° phase and 6 dB gain margin for such controllers. However, for even the single-input, single-output case there has re-

[8] Doyle, John. "Guaranteed margins for LQG regulators." IEEE Transactions on automatic Control 23.4 (1978): 756-757.

H_∞ Control

Consider a closed-loop system under policy π with disturbances $w(t)$ and cost-relevant outputs $z(t)$:

$$\dot{x}(t) = f_\pi(x(t), w(t)), \qquad z(t) = h_\pi(x(t), w(t))$$

$f(x(t), \pi(x(t)), w(t))$

$\|z(t)\|^2 = x^T Q x + u^T R u$

$\begin{bmatrix} Q^{1/2} x \\ R^{1/2} u \end{bmatrix}$

Define a norm over a time-varying function:

$$\|x(\cdot)\|_{L_2} = \sqrt{\int_0^\infty x(t)^\top x(t) dt}$$

The H_∞ problem is then (assume $x(0) = 0$):[9]

$$\min_{\pi \in \Pi} \max_{\|w\|_{L_2} \neq 0} \frac{\|z(\cdot)\|_{L_2}}{\|w(\cdot)\|_{L_2}}$$

HOW DOES A DISTURBANCE AMPLIFY COST?

[9] Zhou, Kemin, John Comstock Doyle, and Keith Glover. Robust and optimal control. Vol. 40. New Jersey: Prentice hall, 1996.

Lyapunov Functions as Certificates in H_∞ Control

Suppose we want to build a certificate of our objective:

$$\max_{\|w\|_{L_2} \neq 0} \frac{\|z(\cdot)\|_{L_2}}{\|w(\cdot)\|_{L_2}} \leq \gamma.$$

This is equivalent to saying the following holds for all $\|w\|_{L_2}$:

$$\|z(\cdot)\|_{L_2}^2 - \gamma^2 \|w(\cdot)\|_{L_2}^2 \leq 0.$$

We can certify this with a Lyapunov function $V(x)$ satisfying:

$$\dot{V}(x) + z(t)^\top z(t) - \gamma^2 w(t)^\top w(t) \leq 0.$$

For all $w(\cdot)$. (and $V(0) = 0$ and $V(x) \geq 0$)

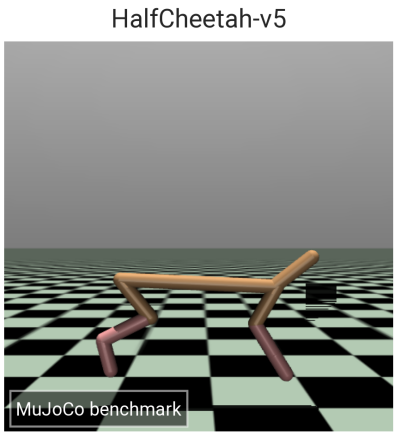
Example: Domain Randomization in RL

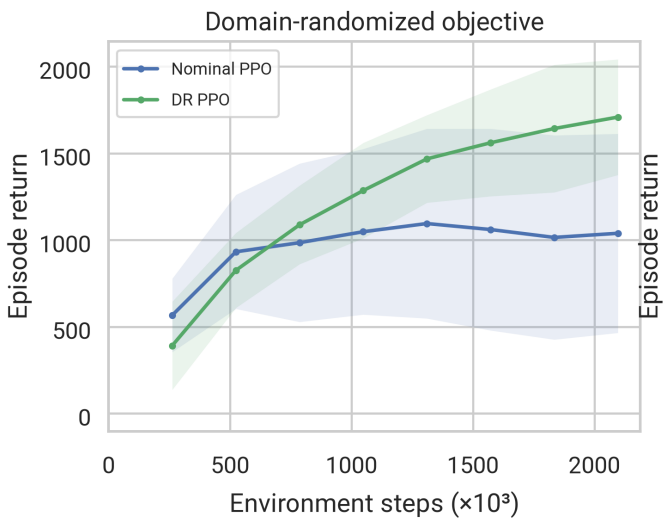
Reinforcement learning with domain randomization tries to minimize some cost:

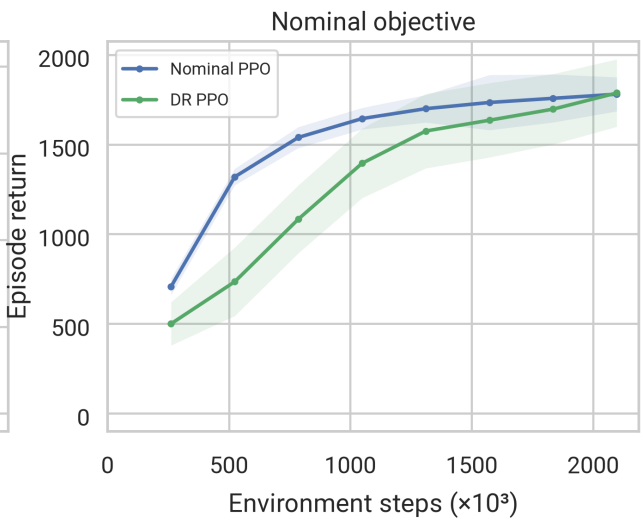
$$J(\theta) = -\mathbb{E}_{\xi}\mathbb{E}_{\tau\sim\pi_\theta,\xi}\left[\sum_{t=1}^{T}\gamma^t r(x_t, u_t)\right]$$

where ξ are random domains (friction, etc.)

Example:







Other Measures of Risk: VaR and CVaR

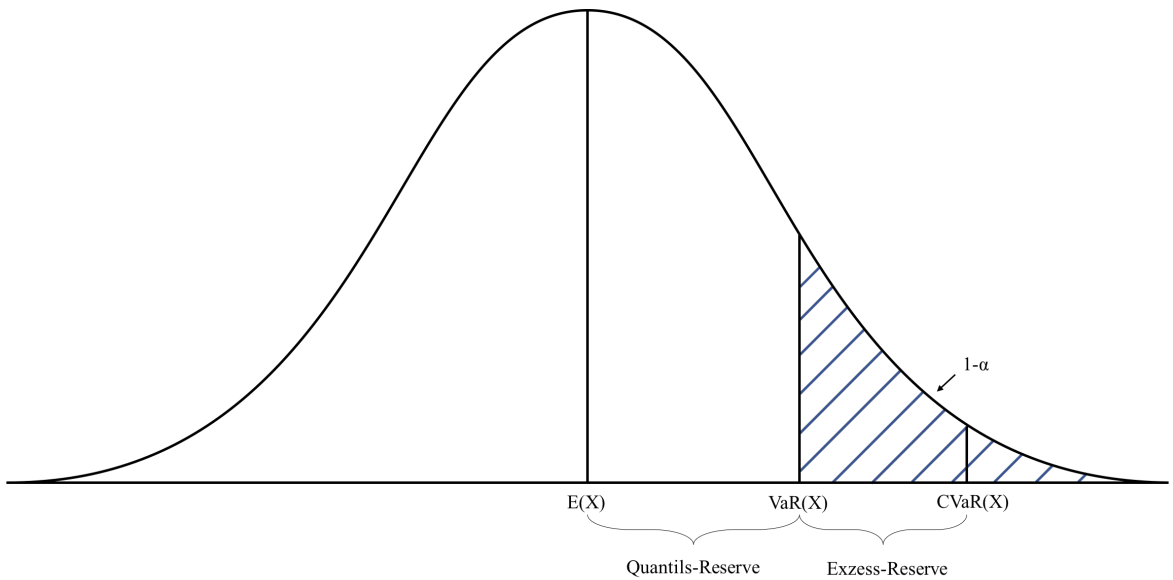
Let J be a random variable denoting some sort of cost (e.g. the trajectory cost over w_t). Maybe we don't want to just focus on $\mathbb{E}J$.

The value-at-risk at level α is a quantile:

$$\mathsf{VaR}_\alpha(J) = \min\{z : \mathbb{P}(J \leq z) \geq \alpha\}$$

The conditional value-at-risk for continuous distributions is the expected tail cost:

$$\mathsf{CVaR}_\alpha(J) = \mathbb{E}[J \mid J \geq \mathsf{VaR}_\alpha(J)]$$



Distributionally Robust Control

We could also strike a balance between worst-case and expectational objectives by minimizing over an *ambiguity set* of distributions:[10]

$$\min_{\pi} \max_{Q \in \mathcal{Q}} \mathbb{E}_Q \left[J(\pi; w) \right]$$

where $\mathcal{Q}$ is a set of probability distributions.

Example: A KL ball:

$$\mathcal{Q} = \left\{ Q \in \Delta^{\mathcal{W}} : D_{\mathsf{KL}}(Q \| P) \leq \epsilon \right\}$$

[10] Rahimian, Hamed, and Sanjay Mehrotra. "Frameworks and results in distributionally robust optimization." Open Journal of Mathematical Optimization 3 (2022): 1-85.

Conclusion

Feedback matters

- In uncertain systems, feedback acts to steer the distribution, preventing over-conservatism.

Stochastic/Robust problems are harder to solve

- More variables; more hyperparameters
- Often non-convex.

There are many methods and techniques for solving

- Linearizing systems with Gaussian noise or Zonotopes
- Sampling
- Tube trajectory optimization
- Handling chance constraints
- etc.